

Calculus II

Integrals of the form $\int \sqrt{ax^2 + c} dx, a, c > 0$

Todor Milev

2019

Euler substitution: $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$, $\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right)$,
 $t = \sqrt{x^2 + 1} - x$, $dx = -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt$. Recall $t > 0$.

Example

$$\begin{aligned}
 \int \sqrt{x^2 + 1} \, dx &= - \int \frac{1}{2} \left(\frac{1}{t} + t \right) \frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\
 &= -\frac{1}{4} \int \left(\frac{1}{t^3} + 2\frac{1}{t} + t \right) dt \\
 &= -\frac{1}{4} \left(-\frac{t^{-2}}{2} + 2 \ln |t| + \frac{t^2}{2} \right) + C \\
 &= \frac{1}{2} \left(\frac{1}{2} \left(t^{-1} - t \right) \frac{1}{2} \left(t^{-1} + t \right) \right) - \frac{1}{2} \ln t + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} - \frac{1}{2} \ln \left(\sqrt{x^2 + 1} - x \right) + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln \frac{\sqrt{x^2 + 1} + x}{\left(\sqrt{x^2 + 1} - x \right) \left(\sqrt{x^2 + 1} + x \right)} + C \\
 &= \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln \left(\sqrt{x^2 + 1} + x \right) + C
 \end{aligned}$$

Example

$$\begin{aligned}
 \int \frac{1}{x^2 \sqrt{x^2 + 9}} dx &= \int \frac{1}{x^2 3 \sqrt{\left(\frac{x}{3}\right)^2 + 1}} dx \\
 &= \int \frac{1}{(3 \cot \theta)^2 3 \sqrt{\cot^2 \theta + 1}} d(3 \cot \theta) \\
 &= \int \frac{1}{27 \cot^2 \theta \sqrt{\csc^2 \theta}} (-3 \csc^2 \theta) d\theta \\
 &= \frac{1}{9} \int \frac{-\csc^2 \theta}{\cot^2 \theta \csc \theta} d\theta \\
 &= \frac{1}{9} \int \frac{-\sin \theta}{\cos^2 \theta} d\theta = \frac{1}{9} \int \frac{1}{\cos^2 \theta} d(\cos \theta)
 \end{aligned}$$

Set

$$\begin{aligned}
 \frac{x}{3} &= \cot \theta \\
 x &= 3 \cot \theta
 \end{aligned}$$

$$\theta \in (0, \pi)$$

$$\begin{aligned}
 \theta &\in (0, \pi) \Rightarrow \\
 \csc \theta &> 0
 \end{aligned}$$

Set $u = \cos \theta$